

Scala Advanced + Spark - Exercise

Data Engineering Training | End-to-End Transformation Lab

Scenario	An e-commerce team receives daily orders with missing fields, inconsistent values and duplicate business activity. Build a Spark/Scala pipeline that converts the raw file into analytics-ready outputs.

Task 1 - Create and Inspect the Dataset

- Create orders.csv with: order_id, customer_id, order_date, product, category, price, quantity, city.
- Include at least: one missing city, one missing product, one missing price, and one duplicate order_id.
- Load the file with Spark and inspect schema, first rows and row count.

Deliverable: orders.csv and inspection output.

Task 2 - Enforce an Explicit Schema

- Define a StructType rather than relying on inferSchema.
- Choose appropriate types for identifiers, price and quantity.
- Explain two production reasons for schema enforcement.

Deliverable: Schema code and printSchema output.

Task 3 - Clean and Standardize

- Remove or quarantine rows with missing product or price.
- Fill missing city with 'Unknown'.
- Remove duplicate order IDs using a documented rule.
- Validate that quantity and price are positive.

Deliverable: Cleaned DataFrame count plus rejected-record count.

Task 4 - Feature Engineering

- Create total_amount = price * quantity.
- Derive order_year and order_month from order_date.
- Create an order_value_band such as Low, Medium, High with thresholds you document.

Deliverable: Sample transformed rows.

Task 5 - Business Metrics

- Calculate total revenue per product.
- Return top 3 cities by revenue.
- Calculate average order value.
- Find the most sold product by quantity.

- Calculate monthly revenue trend.

Deliverable: Spark output for all five metrics.

Task 6 - Advanced Analytics

- Calculate revenue by city and category.
- Calculate customer lifetime value as total spend per customer.
- Return customers whose lifetime value exceeds 1,500.
- Identify customers whose order value is above their own average order value.

Deliverable: Advanced analytics output.

Task 7 - Persist and Validate

- Write the cleaned analytics-ready data to Parquet.
- Read the Parquet output back and validate row count and schema.
- Document what would change if this became an incremental daily production pipeline.

Deliverable: Parquet output + validation + README.

Challenge: Explain how late-arriving data, duplicates and schema changes would be handled.